

PMOS Risk Prediction Using Machine Learning

Abstract

Polyendocrine Metabolic Ovarian Syndrome (PMOS), formerly known as **Polycystic Ovary Syndrome (PCOS)**, is a common endocrine and metabolic disorder affecting women of reproductive age. PMOS is associated with hormonal imbalance, irregular menstrual cycles, weight gain, acne, excessive hair growth, and an increased risk of infertility and metabolic disorders. Early identification of individuals at risk is important for timely medical intervention and improved health outcomes. This project proposes a **PMOS Risk Prediction System** using the **Logistic Regression** machine learning algorithm to predict the risk of PMOS based on health-related features such as **Age, Height, Weight, Body Mass Index (BMI), Menstrual Cycle (Regular/Irregular), Weight Gain, Hair Growth, and Acne**. The dataset is preprocessed by removing duplicate records, handling missing values, and encoding categorical variables before training the model. Random Forest is selected because it is a reliable classification algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Based on the prediction result, the system provides basic health recommendations, including maintaining a balanced diet, engaging in regular physical activity, managing body weight, and consulting a qualified healthcare professional when the predicted risk is high. The proposed system serves as an early screening tool to support awareness and encourage timely medical consultation. However, it is intended only as a decision-support tool and should not replace professional medical diagnosis.

Keywords: PMOS, Logistic Regression, Machine Learning, Classification, Disease Prediction, Healthcare Analytics.

References:

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